NATHAN FREKHTMAN

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LinkedIn

Education

- San Diego State University, San Diego, CA
- B.S. Mechanical Engineering undergraduate, GPA: 3.8

Past Experiences

Mechanical Engineering Intern | Techstars/ZipLaunchpad | San Diego, CA | Nov 2025 - Feb 2026
Selected for a competitive university-sponsored Techstars startup internship program.
Hapware Inc: Led mechanical design of glasses-mounted camera enclosure integrating ESP32 camera AI module, battery, and antenna within a one-week deadline.

- Engineered intuitive snap-fit closures and alignment pegs, delivering a fully functional two-piece FDM prototype for external testing in 4 days.

ORBES Inc: Developed a rigid 80/40 aluminum extrusion frame for an air bearing table, optimizing architecture to maximize stiffness under spatial constraints.

- Compiled and managed a comprehensive Bill of Materials (BOM), successfully navigating vendor procurement for specialized epoxy materials.

Idea Lab: Constructed a sound-activated projectile launcher, integrating an electret mic array with Raspberry Pi Pico and ESP32 to drive motorized targeting.

- Troubleshot electrical shorts and conducted root-cause analysis on a shear failure to design reinforced structural geometries for the V2 iteration.

Mechatronics Club Member | Robosub Team, SDSU | September 2025 - Present

- Engineered an underwater scissor-lift deployment mechanism, iterating through CAD designs to ensure smooth actuation in fluid environments.
- Fabricated and assembled mechanical linkages, collaborating cross-functionally to validate mechanical and software performance during extensive pool testing.

VR Flight Simulator Joystick Adapter | Personal Project | Summer 2024

- Architected a 3-axis gimbal assembly in Fusion 360 to accurately map VR controller movements to realistic pitch, roll, and yaw aircraft dynamics.
- Iterated through multiple rapid FDM prototypes using PLA to perfect custom self-centering cam mechanisms.
- Integrated a custom magnetic coupling interface enabling seamless, tool-less transitions between free-hand VR and rigid joystick modes.

Flight Simulator Rudder Pedals | Personal Project | Summer 2025

- Formulated a fully 3D-printed rudder pedal system utilizing FDM kinematic joints to strictly navigate a zero-purchased-hardware constraint.
- Designed a custom wheel-supported sliding base to replace traditional metal linear rails, reducing estimated manufacturing costs and complexity.
- Evaluated the prototype through physical load testing; identified off-axis instability causing shear failure at a critical pin joint to establish constraints for future revisions.

Skills

- **CAD**: Fusion 360, SolidWorks, Onshape
- **Programming:** Python, Arduino C++, MicroPython
- **Embedded & Tools:** Arduino IDE, PlatformIO, GitHub
- **Hardware:** Microcontrollers, servos, stepper motors
- **Fabrication:** FDM 3D printing, laser cutting, manual fabrication (hand tools, drill press, rotary tools)
- **Other:** Engineering Documentation, Bill of Materials

Interests

Gamedev, hobby programming, flight simulators, light drawing/sketching, and logic/puzzle games.